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Reconstructing Spiking Neural Networks Using a Single Neuron with Autapses

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Abstract

Spiking neural networks (SNNs) are promising for neuromorphic computing, but high-performing models still rely on dense multilayer architectures with substantial communication and state-storage costs. Inspired by autapses, we propose time-delayed autapse SNN (TDA-SNN), a framework that reconstructs SNNs with a single leaky integrate-and-fire neuron and a prototype-learning-based training strategy. By reorganizing internal temporal states, TDA-SNN can realize reservoir, multilayer perceptron, and convolution-like spiking architectures within a unified framework. Experiments on sequential, event-based, and image benchmarks show competitive performance in reservoir and MLP settings, while convolutional results reveal a clear space–time trade-off. Compared with standard SNNs, TDA-SNN greatly reduces neuron count and state memory while increasing per-neuron information capacity, at the cost of additional temporal latency in extreme single-neuron settings. These findings highlight the potential of temporally multiplexed single-neuron models as compact computational units for brain-inspired computing.

1. Introduction

Spiking neural networks (SNNs) are widely regarded as the third generation of neural networks [17, 26]. They serve as an important foundation for neuromorphic computing [31], featuring event-driven processing, high energy efficiency, and rich temporal dynamics [12, 53]. However, most high-performing SNNs still adopt dense multilayer architectures similar to artificial neural networks (ANNs) [19, 32, 41–43]. Such designs require extensive neuron communication and large state storage, limiting spatial efficiency and hindering deployment in resource-constrained settings [48, 49, 51]. This limitation motivates the search for compact SNN architectures that preserve computational capability while reducing structural redundancy. [46]

The intrinsic computational potential of single neurons provides a promising direction for advancing information processing. Biological neurons not only perform dendritic integration [24, 35] but also exhibit complex mechanisms such as intrinsic plasticity [13] and lateral inhibition [28], enabling rich nonlinear dynamics across spatiotemporal dimensions. In the SNN field, various strategies have been proposed to enhance single-neuron expressiveness, including dendritic computation [25], intrinsic plasticity [15, 38], lateral inhibition–inspired competition mechanisms [10, 50], neuronal heterogeneity [30, 52], and multicompartment modeling [9]. Although these approaches improve the nonlinear computational capability of individual neurons to some extent [23], their internal states still rely mainly on instantaneous inputs, which limits their ability to retain long-term information or perform recursive computations [5]. Among these, cerebellar Purkinje cells exemplify single-neuron complexity through elaborate dendrites and autaptic self-feedback, supporting temporal processing and memory.

In exploring the computational potential of single-neuron structures, existing engineering approaches based on single-neuron delay loops offer valuable insights. Studies on single-neuron deep learning [36, 37] and one-core reservoir computing (RC) models [29] demonstrate that a single node possesses significant capacity for temporal information processing. However, these methods typically rely on non-biological hardware, and their continuous-value computations fundamentally differ from spike-based, event-driven mechanisms, limiting their direct applicability to SNNs and biological interpretation.

In biological neural systems, autapses provide a natural mechanism for temporal self-interaction. An autapse is a synaptic connection that a neuron forms with itself, enabling it to sense its own past spiking activity, and to exert either inhibitory or excitatory modulation on its cur-

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rent state [2, 40, 45]. From a computational perspective, autapses introduce intrinsic temporal memory within neurons, naturally embedding the high-dimensional information into their dynamics and providing a biological basis for single neurons to realize long-term dependencies and recursive computation [33, 47]. Such autaptic self-modulation in Purkinje cells inspires our model design for long-term dependencies and recursive computation.

Based on this observation, we propose a leaky integrate-and-fire neuron with delayed autapses (TDA-LIF), which forms the basis of a corresponding spiking neural network model termed TDA-SNN. Under different structural configurations, the TDA-LIF neuron reorganizes its internal temporal-node evolution to construct reservoir-computing, multilayer-perceptron, and convolution-like spiking architectures within a unified framework. Experiments on sequential, event-based, and image datasets show that TDA-SNN provides competitive performance in reservoir and MLP settings, while its convolutional results reveal a clear space–time trade-off. Compared with structurally equivalent standard SNNs, TDA-SNN drastically reduces neuron count and state memory and increases per-neuron information capacity, at the cost of additional temporal latency in extreme single-neuron settings. Our main contributions are summarized as follows:

- We theoretically show that a TDA-LIF neuron can constructively realize three representative SNN structures: reservoir computing (RC), multilayer perceptrons (MLPs), and convolution-like architectures.
- We propose a dedicated optimization framework for TDA-SNN to address the training challenges introduced by internal delayed feedback, enabling stable learning of temporally unfolded single-neuron models.
- Extensive experiments on multiple benchmarks demonstrate that TDA-SNN achieves competitive performance in RC and MLP settings, and reveal a clear space–time trade-off in convolutional settings, while substantially reducing neuron count and state memory and significantly enhancing per-neuron information capacity.

2. Related Works

2.1. Neurons with High Computing Capacity

Early artificial neuron models, such as McCulloch-Pitts neurons [27], greatly simplify the richer dynamics of biological neurons, whose dendrites, ion channels, and nonlinear membrane properties support powerful local computation. [7] In particular, dendrites can integrate inputs, perform nonlinear operations, and even implement logical functions [18, 24, 35], motivating SNN studies that enhance single-neuron capability through dendritic computation, intrinsic plasticity, lateral inhibition, neuronal heterogeneity, and multi-compartment modeling. [9, 10, 15, 23, 25, 30, 38, 50, 52] However, the role of temporal delay in improving single-neuron computation remains underexplored. [3, 5, 8]

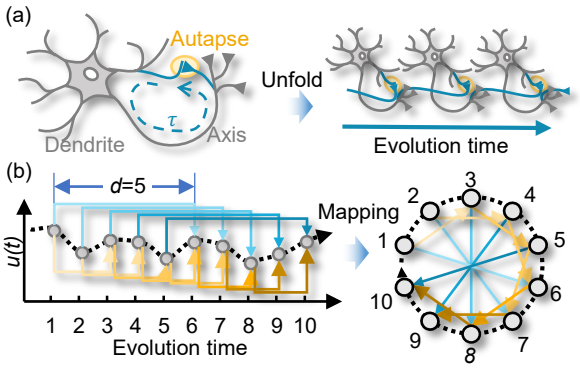


Figure 1. Evolution and folding principles of autapses. (a) An autapse in biological neurons and its evolutionary unfolding. (b) Distribution of evolutionary time in the TDA-LIF model and its mapping to reservoir computation.

2.2. Time-Delay Neural Network

Transmission delay is a fundamental resource for temporal processing in both biological and artificial neural systems. [4, 34] Time-delay neural networks explicitly model temporal dependencies through delayed connections [20, 39], and folded-in-time deep networks further compress depth into delay loops within a single node, achieving strong performance in vision and EEG tasks. [29, 36, 37] Yet these approaches typically rely on continuous-valued signals and have limited biological plausibility. In contrast, biological autapses allow neurons to regulate current activity using their own past spikes. [2, 40, 45] This perspective motivates us to view delay as an intrinsic autaptic property. It enables a single spiking neuron to emulate network-like temporal computation while preserving interpretability and improving long-range temporal modeling.

3. Method

We introduce time-delayed autapses into the LIF neuron to enhance its temporal dynamics and internal memory. Based on this modification, we construct a temporally unfolded single-neuron model and show how it can be mapped to reservoir-computing, MLP, and convolution-like spiking architectures. Unless otherwise stated, t denotes the index of an internal temporal node, while T denotes the external data time window used for sequential or event-based inputs.

3.1. LIF Neuron with Time-delayed Autapses

Introducing an autapse with delay d into the LIF neuron [16] provides temporal self-feedback: a spike emitted by the neuron is fed back to its dendrites after d internal nodes. As shown in Fig. 1, the axon connects to the neuron's own den-

drites, allowing past spikes to influence future membrane-potential evolution.

Mathematically, the iterative formula for the TDA-LIF is as follows:

$$v^t = \tau v^{t-1}(1 - s^{t-1}) + I^t + I^t_{autapse}, \tag{1}$$

where $I^t_{autapse} = \sum_{d \in D_t} w_a^d s^{t-d}$ is the delayed autaptic current generated by the neuron's own past spikes, $D_t = \{d \in \mathbb{N} \mid d < t\}$ is the set of valid delays at node t, and w_a^d denotes the weight of the autapse with delay d.

3.2. Temporal Dynamics to Reservoir Computing

Based on the TDA-LIF model, we unfold neuron dynamics into a sequence of temporal nodes, where autapses connect nodes across different internal timesteps. We refer to each internal timestep as a node to distinguish it from the external data time window. These connections are directional and transmit signals forward along the unfolded trajectory. For simplicity, autapses with the same delay are treated as one delay type. As shown in Fig. 1, the neuron evolves over ten nodes, with blue and yellow autapses representing delays of five and two nodes, respectively.

When autapses with different delays are assigned different weights, delayed self-feedback modulates the membrane-potential trajectory and enriches the internal temporal state. Under a finite set of delays, the membrane dynamics can be written as:

$$v^t = \tau v^{t-1}(1 - s^{t-1}) + Wx^t + \sum_{d \in D} w_a^d(t) s^{t-d}, \tag{2}$$

where x^t is the external input injected into node t, D denotes the set of selected autaptic delays, $w_a^d(t)$ is the weight of the autapse with delay d at node t, and $W \in \mathbb{R}^{N \times N_{\text{in}}}$ represents the weight matrix for the external inputs, with N being the number of nodes and N_{in} the input dimension.

When arranged as in the right panel of Fig. 1(b), the unfolded temporal graph induces an RC-like connectivity pattern. Signals propagate along the nodes following the dotted arrows, while delayed autaptic feedback continuously modulates the hidden dynamics. The resulting interaction across temporal nodes enriches the internal states, and aggregating neuronal responses over the unfolded trajectory yields a high-dimensional representation for reservoir computing.

3.3. MLP Equivalence by Autapse Pruning

As shown in Fig. 2, the unfolded trajectory is divided into two segments, highlighted by the red dashed line, and autapses are pruned accordingly. Specifically, autapses confined within the same segment are removed, while those crossing the segment boundary are retained. Rearranging the temporal nodes into two groups then yields a feedforward network structure.

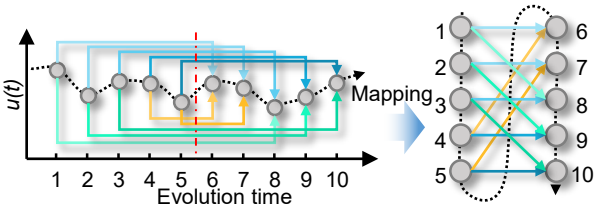


Figure 2. Division of the evolutionary time in the TDA-LIF model and its mapping to an MLP

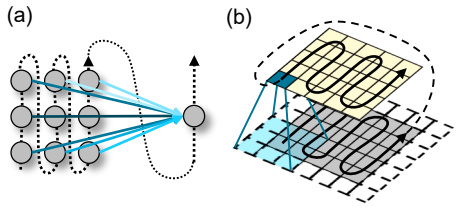


Figure 3. Convolutional layer with the TDA-LIF model. (a) Mapping 10 nodes to a 3×3 convolution. (b) Equivalence of TDA-LIF to a single-channel convolutional layer.

This construction produces an MLP-like structure from a single TDA-LIF neuron. In Fig. 2, the neuron integrates three delayed autaptic groups, $d = 2$ (yellow), 5 (blue), and 7 (green), yielding ten effective synaptic connections between the two temporal segments. During the evolution of the second segment, the dynamics can be written as:

$$v^t = \tau v^{t-1}(1 - s^{t-1}) + Wx^t + \sum_{d \in D_{\text{FC}}} w_a^d(t) s^{t-d}, \tag{3}$$

where $D_{\text{FC}} = \{d \in \mathbb{N} \mid d \le t,\ t - d < t_s\}$ denotes the set of delays that connect the first segment to the second, and t_s is the segment boundary.

Building on this foundation, the TDA-LIF model can be extended to deeper feedforward networks by introducing more temporal segments. Inputs are assigned to the first segment, and spike responses from the final segment are used for recognition. In the corresponding inter-segment connection matrix, the diagonal and parallel-diagonal elements represent autaptic connections with identical delays.

3.4. Convolutional Structure Equivalence

Starting from the MLP-like topology, we assign each spatial output node a set of autapses with shared weights and corresponding delays, thereby encoding the weight-sharing mechanism of a convolution kernel into the neuron's internal temporal structure. These shared autaptic weights play the role of kernel parameters, while distinct delay groups provide a temporal realization of local spatial aggregation.

During signal propagation, output nodes operate in parallel and synchronously, each using the shared autaptic kernel to process a different local region of the input. In this

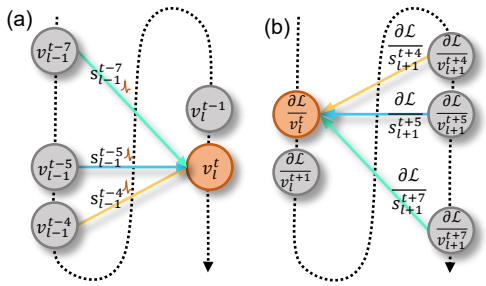


Figure 4. Forward (a) and backward (b) processes in the TDA-SNN model.

way, kernel translation across space is mapped to a repeated delay-sharing pattern across temporal nodes. As illustrated in Fig. 3, this construction provides a convolution-like realization in the spike domain and unifies local spatial aggregation with temporal dynamics.

For example, consider a convolution with a 3×3 kernel, padding 1, and stride 1, with a single input and output channel for simplicity. The TDA-LIF state at a given node is determined by delayed inputs from nine neighboring positions in the previous layer, corresponding to nine delay groups. Because all spatial locations share the same delayed autaptic weights, the unfolded dynamics recover the key properties of local connectivity and weight sharing. This yields a constructive convolution-like mapping, which we later evaluate empirically in a preliminary setting.

3.5. Prototype Learning Method

To effectively decode neuronal firing patterns, it is essential to exploit their intrinsic spatio-temporal dynamics. Conventional static decoding methods neglect temporal depen dencies and fail to align with the spiking nature of SNN outputs. To address this limitation, we introduce a spatiotemporal prototype learning framework that matches output spike sequences to learnable prototypes, enabling efficient and interpretable recognition of dynamic firing behaviors [6]. See the supplementary materials for additional details. To effectively decode neuronal firing patterns, it is essential to exploit their intrinsic spatio-temporal dynamics. Conventional static decoding methods neglect temporal dependencies and fail to align with the spiking nature of SNN outputs. To address this limitation, we introduce a spatiotemporal prototype-learning framework that matches output spike sequences to learnable prototypes, enabling efficient and interpretable recognition of dynamic firing behaviors [6]. Additional implementation details are provided in the supplementary material.

Specifically, a set of binary prototypes $\mathbf{K} \in \{0,1\}^{C\times T}$ is defined to represent target spiking patterns for C classes over T steps. Let $f(x;\theta)$ denote the encoded spatiotemporal representation of input x. The similarity between $f(x;\theta)$ and the i-th prototype $\mathbf{k}_i \in \mathbf{K}$ is measured by

$$d_i(x) = -\left\| f(x;\theta) - \mathbf{k}_i \right\|_2^2 .$$

Based on the distances between the encoded representation and class prototypes, the classification objective is formulated as:

$$\mathcal{L} = -\log \frac{e^{d_i(x)}}{\sum_{j=1}^{C} e^{d_j(x)}} - \lambda d_i(x), \tag{4}$$

where C represents the number of classes, and λ is a hyperparameter controlling the contribution of the prototype regularization term (set to 0.001 in our experiments).

For network optimization, we adopt the spatio-temporal backpropagation (STBP) algorithm, which has been widely used in SNNs due to its efficiency in handling temporal dependencies. [42, 43] To overcome the non-differentiability of spike firing in TDA-LIF neurons and prototype binarization, we replace the gradient with a smooth approximation using the arctangent function:

$$s^t = g\left(v^t\right) \approx \frac{1}{\pi}\arctan\left[\frac{\pi}{2}\alpha(v^t - v_{\text{th}})\right] + \frac{1}{2}. \tag{5}$$

Fig. 4 illustrates information flow in TDA-SNN. In forward propagation, node information arises from both delayed autaptic spikes and the membrane potential carried over from the previous node, as shown in Fig. 4(a). Accordingly, in backward propagation, the gradient at each node is determined by the membrane-potential gradients of its successor node and of the future nodes that receive its delayed spikes, as shown in Fig. 4(b).

For the RC and MLP settings, the membrane-potential gradient at node t is:

$$\begin{aligned}\nabla_{v^t}\mathcal{L} &= \nabla_{v^{t+1}}\mathcal{L}\cdot\frac{\partial v^{t+1}}{\partial v^t} + \sum_{d\in D}\left[\nabla_{v^{t+d}}\mathcal{L}\cdot\frac{\partial v^{t+d}}{\partial v^t}\right] \\ &= \nabla_{v^{t+1}}\mathcal{L}\cdot\tau(1-s^{t-1}) \\ &\quad + \sum_{d\in D}\left[\nabla_{v^{t+d}}\mathcal{L}\cdot w_a^d\cdot\frac{\partial s^t}{\partial v^t}\right]\end{aligned}, \tag{6}$$

where w_a^d is the autaptic weight connecting the spike at node t to the future node $t+d$. The corresponding weight gradients are

$$\begin{aligned}\nabla_w\mathcal{L} &= \nabla_{v^t}\mathcal{L}\cdot x^t, \\ \nabla_{w_a^d}\mathcal{L} &= \nabla_{v^{t+d}}\mathcal{L}\cdot s^t.\end{aligned} \tag{7}$$

Using equations (4)–(7), TDA-SNN can be trained end-to-end with surrogate gradients.

4. Experiments

4.1. Experimental Setup

We evaluate TDA-SNN across multiple benchmarks under different structural mappings. DEAP [21] and SHD [11]

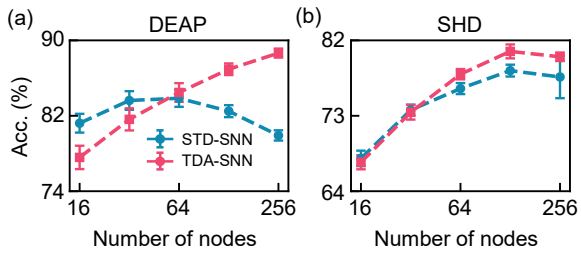


Figure 5. Comparison of STD-SNN and TDA-SNN performance under the RC structure with different reservoir sizes on (a) DEAP and (b) SHD datasets.

are used to evaluate the reservoir-computing setting; MNIST [14], Fashion-MNIST (fMNIST) [44], and DVS Gesture [1] are used to validate the MLP setting; and DVS Gesture together with CIFAR10 [22] are further used to study the convolution-like setting. All models are implemented in PyTorch and trained on an NVIDIA A100 GPU using the Adam optimizer with a cosine learning-rate decay schedule. Unless otherwise stated, training is conducted for 100 epochs. For RC and MLP experiments, results are averaged over 10 independent runs and reported as mean $\pm$ standard deviation; due to the higher computational cost of convolutional experiments, those results are averaged over 5 runs. To ensure a fair comparison, TDA-SNN and the corresponding standard SNN baselines use aligned training protocols and matched architectural scales for each setting. The complete hyperparameter configurations are provided in Supplementary Sec. 9.

4.2. Reservoir Computing Equivalence

To assess the RC capability of TDA-SNN, we conducted experiments on the DEAP and SHD datasets. We further investigate how reservoir size and autaptic selection strategies affect connectivity patterns and classification performance.

4.2.1. Analysis of RC Equivalence

To compare the performance of the standard SNN (STD-SNN) and the proposed TDA-SNN with varying reservoir sizes, we conducted a series of experiments. Specifically, the number of reservoir neurons in the SNN was matched to the number of internal evolution nodes in the TDA-SNN to ensure comparable model complexity, thereby enabling a fair assessment of their representational and learning capabilities. As the reservoir size increased, the projection space became richer and more structured, leading to improved feature separability and discriminative performance. In this ablation study, reservoir sizes of 16, 32, 64, 128, and 256 were evaluated, with fully delayed autapses employed in all configurations.

As shown in Figs. 5(a) and (b), TDA-SNN underperforms STD-SNN at small reservoir sizes (e.g., 16 nodes), reaching 77.59$\pm$1.24% versus 81.21$\pm$1.00% on DEAP and 67.45$\pm$0.84% versus 67.92$\pm$0.92% on SHD. As the reservoir size increases, TDA-SNN improves steadily and becomes competitive with, or even surpasses, STD-SNN at larger sizes. At 256 nodes, TDA-SNN reaches 88.65$\pm$0.48% on DEAP and 80.04$\pm$0.51% on SHD, while STD-SNN attains 79.92$\pm$0.54% and 77.63$\pm$2.53%, respectively. On SHD, the best TDA-SNN result is obtained at 128 nodes (80.68$\pm$0.83%), suggesting that excessive internal evolution can eventually introduce redundant temporal encoding. Overall, these results indicate that delayed autaptic dynamics become increasingly effective as the unfolded reservoir grows, making TDA-SNN a competitive alternative to RC-style SNNs in medium- and large-scale settings. As shown in Figs. 5(a) and (b), TDA-SNN underperforms STD-SNN at small reservoir sizes (e.g., 16 nodes), reaching 77.59$\pm$1.24% versus 81.21$\pm$1.00% on DEAP and 67.45$\pm$0.84% versus 67.92$\pm$0.92% on SHD. As the reservoir size increases, TDA-SNN improves steadily and becomes competitive with, or even surpasses, STD-SNN at larger sizes. At 256 nodes, TDA-SNN reaches 88.65$\pm$0.48% on DEAP and 80.04$\pm$0.51% on SHD, while STD-SNN attains 79.92$\pm$0.54% and 77.63$\pm$2.53%, respectively. On SHD, the best TDA-SNN result is obtained at 128 nodes (80.68$\pm$0.83%), suggesting that excessive internal evolution can eventually introduce redundant temporal encoding. These results indicate that delayed autaptic dynamics become increasingly effective as the unfolded reservoir grows, making TDA-SNN a competitive alternative to RC-style SNNs in medium- and large-scale settings.

4.2.2. Effect of autaptic selection strategy

This subsection investigates the influence of autaptic selection strategies on the performance of TDA-SNN under the RC framework. The reservoir size was fixed at 128 neurons, and two strategies, random delay (RD) and maximum connection (MC), were compared across different numbers of delayed autapses (1, 2, 4, 8, 16, 32, and 64).

Table 1 summarizes the classification performance of TDA-SNN under different autaptic selection strategies on DEAP and SHD. On DEAP, both strategies benefit from increasing the number of delayed autapses, and MC shows a slight advantage at larger delay counts (32 and 64), indicating that denser structured delay assignment can improve information propagation when the task is relatively stable. On SHD, the trend is less monotonic: MC drops markedly at two delays, then recovers as more delayed connections are introduced. This observation suggests that on temporally richer datasets, overly constrained delay patterns may be suboptimal in low-delay regimes, whereas sufficient delayed connections allow structured feedback to recover its representational advantage. Thus, the ablation indicates that both the number of autapses and the delay-selection strategy materially affect RC performance, and that moderate-

Table 1. Experimental Results of Autaptic Selection Strategies Across Different Datasets

Dataset	Delay	1	2	4	8	16	32	64
DEAP	MC	83.52±0.52	83.10±1.10	83.63±0.73	83.91±1.14	83.85±0.59	**84.66±0.62**	84.47±0.99
	RD	83.62±1.12	83.34±1.14	83.48±0.64	83.73±1.02	83.63±0.76	83.98±0.81	**84.47±0.99**
SHD	MC	78.35±0.61	72.09±1.17	73.59±0.67	74.69±1.28	77.07±0.48	79.36±0.31	**80.42±0.39**
	RD	78.55±0.68	78.62±0.46	77.60±2.49	78.58±0.25	77.63±1.87	79.26±1.21	**80.42±0.39**
MNIST	MC	33.53±3.51	**94.44±0.10**	94.39±0.16	94.33±0.13	94.37±0.15	94.42±0.20	94.38±0.12
	RD	84.03±19.14	93.32±1.19	94.82±0.15	95.11±0.21	95.63±0.14	95.98±0.11	**96.42±0.09**
	T Inv.	90.33±4.94	93.21±1.84	93.93±0.47	94.37±0.24	94.37±0.20	94.42±0.17	**94.72±0.14**
fMNIST	MC	34.25±2.78	**86.63±0.11**	86.59±0.16	86.60±0.25	86.58±0.18	86.63±0.15	86.55±0.17
	RD	80.51±14.05	85.68±1.28	86.54±0.22	86.88±0.14	87.23±0.13	87.40±0.23	**87.55±0.16**
	T Inv.	83.62±3.30	85.90±0.86	86.32±0.24	86.53±0.21	86.62±0.20	86.59±0.16	**86.74±0.19**
DVS Gesture	MC	23.13±2.86	52.88±1.82	52.71±1.67	52.88±2.32	**55.28±2.12**	54.44±3.10	53.19±2.64
	RD	45.45±8.79	51.84±5.32	53.61±1.91	55.31±2.92	57.88±1.94	58.40±1.69	**59.55±2.04**
	T Inv.	40.87±8.15	50.21±6.71	54.34±2.02	54.79±3.31	55.38±2.35	55.42±2.68	**57.01±2.48**

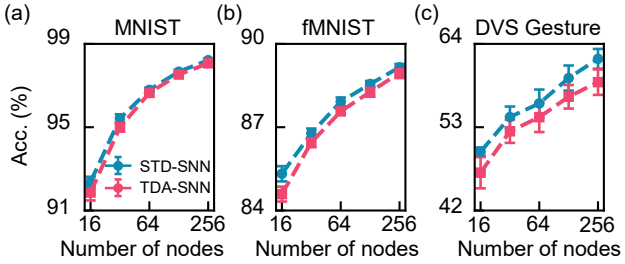


Figure 6. Comparison of TDA-SNN and STD-SNN accuracy under MLP architectures with varying numbers of nodes on (a) MNIST, (b) fMNIST, and (c) DVS Gesture.

to-large delay sets are generally more reliable than sparse configurations. Statistical significance analyses of the experimental results are provided in Supplementary Sec. 11.

4.3. Multilayer Perceptron Equivalence

To further evaluate the generality of TDA-SNN in feedforward settings, we extended the experiments to MLP-like architectures. Specifically, we investigated whether the time-delayed autaptic mechanism can reproduce feedforward spiking representations through temporal unfolding. Following the RC analysis, we examined the effects of temporal-node number, autaptic selection strategy, and network depth on MNIST, fMNIST, and DVS Gesture.

4.3.1. MLP Structural Equivalence

To validate the structural equivalence and representational capability of TDA-SNNs in multilayer feedforward architectures, we compared their performance with STD-SNNs under identical MLP configurations. The network comprises two layers: the first layer performs preliminary feature extraction from external inputs, while the second layer generates the output patterns for classification. All experiments employed fully delayed self-synapses to ensure maximal temporal connectivity. To investigate the effect of network scale, we varied the number of nodes in each layer (16, 32, 64, 128, and 256).

Fig. 6(a)–(c) presents the classification accuracy of STD-SNNs (red) and TDA-SNNs (blue) across different node numbers. TDA-SNN accuracy increases steadily with network size, from 92.37±0.24% to 98.23±0.09% on MNIST, from 85.23±0.27% to 89.16±0.12% on fMNIST, and from 53.99±0.86% to 72.95±2.02% on DVS Gesture. At small scales (e.g., 16 nodes), STD-SNN slightly outperforms TDA-SNN, suggesting that limited representational capacity reduces the benefit of temporal multiplexing. As the number of nodes increases, the gap narrows substantially, indicating that TDA-SNN can serve as a competitive feedforward alternative when sufficient temporal nodes are available. These results support the effectiveness of the proposed temporal construction in medium- and large-scale MLP settings, while also showing that its benefits are less pronounced in low-capacity regimes.

4.3.2. Effect of Strategy Selection

This subsection investigates the impact of autapse delay selection strategies on TDA-SNN performance in MLP architectures. We compare RD, MC, and time-invariant (T Inv.) strategies across varying numbers of delayed autapses, where the T Inv. strategy assigns identical weights to autapses with the same delay. These experiments evaluate how the choice of delay strategy and weight sharing influences temporal encoding and feature extraction.

Increasing the number of autapses improves classification accuracy across MNIST, fMNIST, and DVS Gesture, as

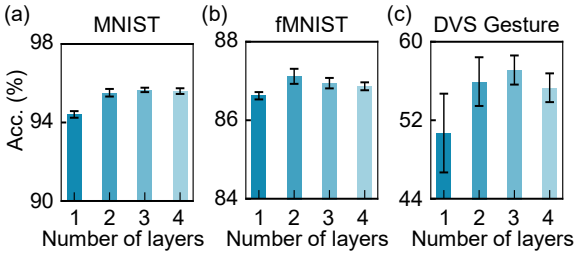


Figure 7. Experimental results of networks with different layer depths on three benchmark datasets. (a) MNIST, (b) fMNIST, and (c) DVS Gesture.

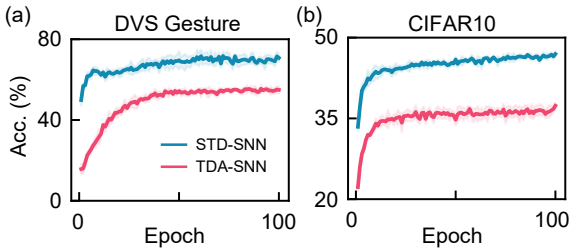


Figure 8. Test accuracy comparison between the TDA-SNN and the STD-SNN under convolutional structure on two benchmark datasets. (a) DVS Gesture and (b) CIFAR-10.

shown in Tab. 1. On MNIST and fMNIST, the three strategies converge quickly once two or more autapses are used, with MC showing a clear disadvantage only in the extreme low-delay regime. On the more temporally demanding DVS Gesture dataset, RD and T-Inv. outperform MC at very small autapse counts, indicating that greater delay diversity is beneficial when temporal structure is complex. As the number of autapses increases beyond four, the gap between strategies narrows and MC becomes comparable to the other two strategies. This ablation suggests that the number of delayed connections is the dominant factor, while the specific selection strategy mainly affects low-resource settings. Statistical significance analyses of the experimental results are provided in Supplementary Sec. 11.

4.3.3. Impact of Layer Number in MLP Structures

We investigated the effect of network depth (N_{layer}) on TDA-SNN performance in MLP structures, fixing 64 nodes per layer and employing the RD selection strategy. Classification performance was evaluated across networks with one to four layers. The results are presented in Fig. 7.

On MNIST, accuracy steadily increases from 94.41 $\pm$ 0.17% to 95.60 $\pm$ 0.15% as depth grows from 1 to 4 layers, indicating that additional depth remains beneficial in this setting. For fMNIST, accuracy rises from 86.62 $\pm$ 0.09% to 87.12 $\pm$ 0.19% at 1–2 layers but then decreases slightly at 3 and 4 layers, suggesting that moderate depth is sufficient. On DVS Gesture, performance improves from 1 to 2 layers and then largely saturates, reaching its best value of 57.12 $\pm$ 1.48% at 3 layers. Thus, the effect of depth is dataset-dependent: deeper networks help on MNIST, moderate depth is adequate on fMNIST, and the benefit saturates early on DVS Gesture. Temporal unfolding supports deeper feedforward constructions, but the optimal depth depends on task complexity and data characteristics.

Further evidence for the robustness of the MLP setting is provided in Supplementary Sec. 13. The MNIST learning-rate results confirm stable optimization, and the CIFAR100 results with different numbers of nodes show that TDA-SNN remains competitive on a more challenging dataset.

4.4. Convolutional Equivalence Exploration

To examine whether the proposed temporal construction can extend beyond RC and MLP settings, we performed preliminary experiments with convolution-like architectures on DVS Gesture and CIFAR-10, which represent dynamic and static visual inputs, respectively. These experiments are intended as an initial proof of concept for applying TDA within convolutional layers. Further implementation details are provided in Supplementary Sec. 14.

Fig. 8 shows classification performance across training epochs. On DVS Gesture, STD-SNN reaches 73.61 $\pm$ 1.93% accuracy and stabilizes after approximately 60 epochs, whereas TDA-SNN converges more slowly and plateaus at 57.58 $\pm$ 1.00%. A similar trend is observed on CIFAR-10, where STD-SNN reaches 47.31 $\pm$ 0.24% and TDA-SNN stabilizes at 37.64 $\pm$ 0.65%. These results suggest that, although delayed self-feedback enriches temporal dynamics, it also introduces substantial optimization difficulty in convolutional settings, likely because delayed feedback interferes with hierarchical spatiotemporal feature aggregation. Therefore, the current single-neuron convolution-like construction should be viewed as a preliminary exploration rather than a fully competitive convolutional alternative. Improving this setting will likely require multi-neuron parallelization to better balance spatial representation and temporal recursion.

4.5. Computational Complexity Analysis

To evaluate efficiency, we analyzed the per-layer complexity of both STD-SNNs and TDA-SNN under three representative architectures: RC, MLP, and convolutional networks. For the RC structure, we used SHD; for MLP, MNIST; and for convolutional networks, CIFAR-10. Both the RC and MLP structures were scaled to a size of 64. We report FLOPs, parameter count, neuron count, and training speed (seconds per epoch). In all cases, TDA-SNN replaces conventional synapses with time-delayed autapses and uses all available autapse delays.

As shown in Table 2, STD-SNNs exhibit high spatial

Table 2. Per-layer computational complexity, parameter count, neuron count, speed, and per-neuron information content across different models.

Model	Arch.	FLOPs	Params.	$N_{\text{Neu.}}$	Speed (s/epoch)	$\mathbf{S}$ (bit)
STD-SNN	RC	N^2T	N^2	N	22.57	489.91
	MLP	$N_{\text{in}}N_{\text{out}}T$	$N_{\text{in}}N_{\text{out}}$	N_{out}	8.36	3113.89
	Conv.	$2C_{\text{in}}C_{\text{out}}k^2WHT$	$C_{\text{in}}C_{\text{out}}k^2$	$C_{out}WH$	17.21	5.48
TDA-SNN	RC	$\sum_d (N-d)\,T$	$\sum_d (N-d)$	1	21.14	32096.37
	MLP	$\sum_d N_d T$	$\sum_d N_d$	1	90.91	199237.63
	Conv.	$2C_{\text{in}}C_{\text{out}}\left(\sum_d N_d\right)WHT$	$C_{\text{in}}C_{\text{out}}\left(\sum_d N_d\right)$	1	3060.23	56486.07

complexity, with RC and MLP structures scaling quadratically or linearly with the number of connections (N^2T and $N_{\text{in}}N_{\text{out}}T$), whereas TDA-SNN converts these spatial couplings into temporal dependencies ($\sum_d N_d T$). This design drastically reduces the number of physical neurons per layer, but the benefit comes at the cost of additional temporal computation due to delayed feedback. The trade-off is mild in the RC setting, where training speed remains comparable, but becomes much more pronounced in MLP and especially convolutional settings. Therefore, the main efficiency advantage of TDA-SNN lies in spatial compactness and neuron-count reduction rather than uniformly lower runtime cost. This observation is consistent with the space–time trade-off discussed throughout the paper.

4.6. Discussion on Single-Neuron Storage Capacity

We further analyzed the storage capacity of individual neurons by quantifying the per-neuron information content under the same experimental settings as in the previous section for RC, MLP, and convolutional networks. Here, $\mathbf{S}$ denotes the effective information content carried by a single neuron and is computed as $\mathbf{S} = \text{Acc.} \times N_{\text{samples}} \times \log_2 C / N_{\text{Neu.}}$. As shown in Table 2, TDA-SNN achieves substantially higher per-neuron information content than STD-SNN across all three architectures. This increase is mainly due to the extreme reduction in neuron count, from N or N_{out} in STD-SNN to 1 in TDA-SNN, while preserving useful predictive performance. In the RC and MLP settings, delayed autapses and temporal integration allow a single neuron to encode information that would otherwise be distributed across an entire layer, leading to a dramatic increase in bits per neuron. In the convolution-like setting, the gain is smaller than in RC and MLP because performance is currently more limited, but it still remains far above that of STD-SNN. These results highlight temporal multiplexing as an effective mechanism for increasing information density at the single-neuron level.

4.7. Parallelization and Space–Time Trade-off

We further examine whether the latency overhead observed in the extreme single-neuron setting can be alleviated by parallelization. Additional results on CIFAR-10 show that the high temporal overhead is specific to the most aggressive compression regime and that TDA-SNN provides a flexible space–time trade-off when more parallel neurons are introduced. As the number of parallel neurons increases, both training and inference time decrease substantially; relative to STD-SNN on CIFAR-10, the training-time overhead drops from 178× to 46×, and with 512 parallel neurons the inference time reaches 0.92× of STD-SNN. These observations indicate that the single-neuron model should be viewed as a limit case for maximizing spatial compactness, while parallel multi-neuron realizations provide more practical operating points for deployment. From a memory perspective, this compact formulation remains appealing: on CIFAR-10, the SNN state memory is reduced from 8 KB in STD-SNN to 4 Bytes in the single-neuron TDA-SNN. Additional visualizations and scalability results are provided in Supplementary Sec. 15.

5. Discussion and Conclusion

Our study shows that a single TDA-LIF neuron can realize core SNN architectures, including RC, MLP, and convolution-like layers, highlighting the computational potential of individual neurons. Inspired by temporal self-modulation in cerebellar Purkinje cells, TDA-SNN converts temporal feedback into compact spiking computation. Experiments show competitive performance in RC and MLP settings, while convolutional results and parallelization analysis reveal a clear space–time trade-off. TDA-SNN also reduces neuron count and increases per-neuron information content. However, the current single-neuron convolution-like setting remains limited in efficiency and accuracy. Future work will extend this framework to multi-neuron settings and adaptive delay mechanisms.

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